

Physics 101 Final Exam (Clean Version with Answer Keys)

1. **A 4.00-kg particle moves along the x axis. Its position varies with time according to $x = t + 2t^3$, where x is in meters and t is in seconds. The kinetic energy of the particle at time $t = 2$ s is:**
 - a. 6050 J
 - b. 50 J
 - c. 162 J
 - d. 1250 J
 - e. 722 J (Correct Answer: e)
2. **A 0.05 kg bead slides without friction around a loop. The bead is released from rest at height $h = 0.41$ m. The radius of the loop is 0.1 m. The speed of the bead (in m/s) at point A:**
 - a. 3.0 (Correct Answer: a)
 - b. 0.0
 - c. 2.0
 - d. 3.5
 - e. 8.0
3. **Two vectors $A = 2i + bj$ and $B = ai + 2j$. If $A - 3B = 0$; a and b are constants, then the angle (in degrees) enclosed between the two vectors is:**
 - a. 90
 - b. 0 (Correct Answer: b)
 - c. None is correct
 - d. 180
 - e. 45
4. **A wheel initially has an angular velocity of 18 rad/s but it is slowing at a rate of 2.0 rad/s^2 . By the time it stops it will have turned through:**
 - a. 410 rad
 - b. 245 rad
 - c. 330 rad
 - d. 81 rad (Correct Answer: d)
 - e. 160 rad
5. **A 2kg block is attached to a vertical spring of force constant 10^3 N/m . The speed as it moves through the equilibrium position is 5 m/s, the kinetic energy (in J) of the block after it moves 5cm beyond the equilibrium position is: (Neglect the effect of gravity and air resistance).**
 - a. 5
 - b. 14
 - c. 20
 - d. 24 (Correct Answer: d)
 - e. 44
6. **At time $t = 0$, a car has a velocity of 16 m/s. It slows down with an acceleration given by $a = -0.50t$, in m/s^2 for t in seconds. At the end of 4.0 s it has traveled:**
 - a. 12 m

- b. 25 m
 - c. 59 m
 - d. 0
 - e. 14 m (Correct Answer: e)
7. **At an instant when a particle of mass 80 g has a velocity of 25 m/s in the positive y direction, a 75-g particle has a velocity of 20 m/s in the positive x direction. What is the speed of the center of mass of this two-particle system at this instant?**
- a. 10 m/s
 - b. 21 m/s (Correct Answer: b)
 - c. 45 m/s
 - d. 36 m/s
 - e. 16 m/s
8. **A 1-N pendulum bob is held at an angle θ from the vertical by a 2-N horizontal force F as shown. The tension in the string supporting the pendulum bob (in newtons) is:**
- a. 1
 - b. Square root of 5 (Correct Answer: b)
 - c. $\cos(\theta)$
 - d. $2/\cos(\theta)$
 - e. $\sin(\theta)$
9. **A 6kg object travels along the x-axis with a velocity of $(6t - 3)$ m/s, then the power (in watts) given to this object at $t = 2.0$ s is:**
- a. 200
 - b. 540
 - c. 410
 - d. 144
 - e. 324 (Correct Answer: e)
10. **A vector A is added to $B = 6i - 8j$. The resultant vector is in the positive x direction and has a magnitude equal to that of A . What is the direction of A ?**
- a. -62 degrees
 - b. 100 degrees
 - c. 74 degrees (Correct Answer: c)
 - d. -81 degrees
 - e. 106 degrees
11. **Block A, with a mass of 50 kg, rests on a horizontal table top. The coefficient of static friction is 0.40. A horizontal string is attached to A and passes over a massless, frictionless pulley as shown. The smallest mass m_B of block B, that will start A moving is:**
- a. 20 kg (Correct Answer: a)
 - b. 70 kg
 - c. 30 kg
 - d. 50 kg
 - e. 40 kg
12. **A potential-energy function for a two-dimensional force is of the form $U(x, y) = 3x^5y - 4x$. The force that acts at the point $(1, 0)$ is:**
- a. $F = -11i$
 - b. $F = -3j$
 - c. $F = 4i$

- d. $F = -11\mathbf{i} - 3\mathbf{j}$ (Correct Answer: d)
 e. $F = 4\mathbf{i} - 3\mathbf{j}$
13. **A person weighing 0.70 kN rides in an elevator that has an upward acceleration of 1.5 m/s^2 . What is the magnitude of the force of the elevator floor on the person?**
 a. 0.81 kN (Correct Answer: a)
 b. 0.11 kN
 c. 0.59 kN
 d. 0.64 kN
 e. 0.70 kN
14. **Displacement can be obtained from:**
 a. the slope of an acceleration-time graph
 b. the slope of an acceleration-time graph
 c. the area under an acceleration-time graph
 d. the slope of a velocity-time graph
 e. the area under a velocity-time graph (Correct Answer: e)
15. **In a given displacement of a particle, its kinetic energy increases by 25 J while its potential energy decreases by 10 J. Determine the work of the nonconservative forces acting on the particle during this displacement.**
 a. +35 J (Correct Answer: a)
 b. +15 J
 c. +55 J
 d. -15 J
 e. -35 J
16. **A ball of mass 0.4 kg is freely dropped from a height of $h_1 = 10\text{m}$, it collides with the ground and rebounds to a maximum height of $h_2 = 8\text{m}$. The impulse (in kg.m/s) due to the collision is:**
 a. 29.5
 b. 10.7 (Correct Answer: b)
 c. 5.6
 d. 1.45
 e. 16.1
17. **A body that moves with a constant speed in a circle:**
 a. has no resultant force acting on it.
 b. has no work done on it. (Correct Answer: b)
 c. is described by all of these.
 d. undergoes no change in velocity.
 e. experiences no acceleration.
18. **The y-coordinate of the center of mass of the three-particle system shown below are:**
 a. 0
 b. 2.5 m
 c. 1.9 m
 d. 1.7 m
 e. 1.4 m (Correct Answer: e)
19. **The figure shows a cylinder of radius 0.7m rotating about its axis at 10 rad/s. The speed of the point P is:**
 a. none of these
 b. 7.0 rad/s

- c. 0.70 m/s
 - d. 7.0 m/s (Correct Answer: d)
 - e. 14 rad/s
20. Calculate the impulse by the force, as shown in the figure below.
- a. 1.0 mN.s
 - b. 10.0 mN.s
 - c. 4.0 mN.s
 - d. 2.0 mN.s
 - e. 40.0 mN.s (Correct Answer: e)
21. The angular position of a point on a wheel is described by $\theta(t) = 5 + 10t + 2t^2$ rad. The angular acceleration of the point at $t = 3$ s in (rad/s^2) is:
- a. 4
 - b. 2
 - c. 6 (Correct Answer: c)
 - d. 5
 - e. 5
22. A particle moves at a constant speed in a circular path with a radius of 2.06 cm. If the particle makes four revolutions each second, what is the magnitude of its acceleration (in m/s^2)?
- a. 15
 - b. 13 (Correct Answer: b)
 - c. 18
 - d. 0
 - e. 20

Part II: Problems

- **Q1:** The blocks shown are released from rest with the spring unstretched. The pulley and the horizontal surface are frictionless. If $k = 200 \text{ N/m}$ and $M = 4.5 \text{ kg}$. Calculate the maximum extension of the spring.
- **Q2:** A 4.2-kg object, initially at rest, "explodes" into three objects of equal mass. Two of these are determined to have velocities of equal magnitudes (5.0 m/s) with directions that differ by 90 degrees. How much kinetic energy was released in the explosion?